# Module IV — Magnetic & dielectric properties

Theory notes: concepts, results and what every symbol means

## Types of magnetism

IN ONE LINE Every solid is magnetic in some way. What kind depends on whether its atoms have **permanent moments** (unpaired electrons) and on how neighbouring moments line up: not at all, randomly, parallel, antiparallel and equal, antiparallel and unequal.

### Magnetic terminology

Put a solid in a magnetic field and it becomes magnetised: tiny current loops inside it line up, or are created, and add their own field. Three quantities describe this. The applied **field strength** $H$ (A/m) is set by the coils outside. The **magnetisation** $M$ (also A/m) is the magnetic moment per unit volume the solid develops. The **flux density** $B$ (tesla) is the total, field plus material. The **susceptibility** measures how strongly the solid responds:

$$\chi = \frac{M}{H}$$

WHERE

- $\chi$ magnetic susceptibility (no units)
- $M$ magnetisation: magnetic moment per unit volume
- $H$ magnetising field

In SI, $M$ and $H$ are both in A/m, so $\chi$ is dimensionless (in general a tensor; a scalar for isotropic media). Per gram-molecule it is the *molar* susceptibility $\chi_m$.

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$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1+\chi)\mathbf{H} = \mu\mathbf{H} = \mu_0\mu_r\mathbf{H}, \qquad \mu_r = 1 + \chi$$

WHERE

- $\mathbf{B}$ magnetic flux density (tesla)
- $\mu_0$ permeability of free space, $4\pi \times 10^{-7}$ H/m
- $\mathbf{H}$ magnetising field
- $\mathbf{M}$ magnetisation
- $\chi$ susceptibility
- $\mu$ permeability of the material
- $\mu_r$ relative permeability, $1 + \chi$

$\mu_0 = 4\pi \times 10^{-7}$ H/m. $B$ in Wb/$\mathbf{m}^2$ (tesla). In vacuum $M = 0$, $\chi = 0$, $\mu_r = 1$, $B = \mu_0 H$.

### Where an atomic moment comes from

Moving charges are currents, and a current loop is a magnet. An atom has three sources of moment:

- **Orbital motion** of the electrons: an electron going round a nucleus is a tiny loop current, with moment $\boldsymbol{\mu}_l = -\dfrac{e}{2m}\mathbf{L}$.

- **Spin** of the electrons: $\boldsymbol{\mu_s = -g\dfrac{e}{2m}\mathbf{S}}$ with $g \approx 2$, so one spin-$\frac{1}{2}$ electron carries about one Bohr magneton.
- **Nuclear spin**: smaller by about $m/M_p \approx 1/1836$, ignored here.

Filled shells cancel in pairs (every spin up has a partner down, every orbit has a counter-rotating twin), so the moment of an atom comes from its **unpaired electrons**. The natural unit is the Bohr magneton,

$$\mu_B = \frac{e\hbar}{2m} = 9.27 \times 10^{-24}\ \mathrm{A\,m^2}\ (\mathrm{J\,T^{-1}})$$

WHERE

| | |
|---|---|
| $\mu_B$ | Bohr magneton, the natural unit of atomic magnetic moment |
| $e$ | electron charge |
| $\hbar$ | reduced Planck constant |
| $m$ | electron mass |

Fe has four unpaired 3d electrons, $\mathrm{Mn^{2+}}$ five, so their moments are a few $\mu_B$. Atoms with only closed shells (inert gases, $\mathrm{Cu^{+}}$, $\mathrm{Cl^{-}}$) have none.

### The five types




Pick a type, then switch the field on. Diamagnets have no arrows until a field creates small opposing ones. Paramagnets have arrows that point anywhere; a field only tilts them a little. In the three ordered types the arrangement is there *without* a field, held by the exchange interaction between neighbours.

| Type | $\chi$ | Temperature law | Origin | Examples |
|---|---|---|---|---|
| **Diamagnetic** | small, **negative** ($\sim -10^{-5}$) | independent of $T$ | orbits change to oppose $B$ (Lenz); no permanent moments | Bi, Cu, water, inert gases |
| **Paramagnetic** | small, positive ($10^{-5}$–$10^{-3}$) | $\chi = C/T$ | permanent moments, randomly oriented | Al, Pt, $O_2$, rare-earth salts |
| **Ferromagnetic** | very large, positive | $\chi = C/(T - T_c)$ above $T_c$ | neighbours parallel (exchange); domains | Fe, Co, Ni, Gd, Dy, MnBi, $CrO_2$ |
| **Antiferromagnetic** | small, positive | $\chi = 2C/(T + \theta)$ above $T_N$, cusp at $T_N$ | neighbours antiparallel, equal | MnO, FeO, NiO, Cr |
| **Ferrimagnetic** | large | ferro-like, Curie point | antiparallel, unequal | ferrites, $Fe_3O_4$ |

Above a critical temperature the differences among the ordered types disappear and every magnetic material becomes paramagnetic: heat scrambles the arrangement.

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**How to tell them apart from data.** Sign of $\chi$: negative means diamagnetic. Size: $10^{-5}$–$10^{-3}$ means dia- or paramagnetic, $10^{2}$–$10^{5}$ means ferro- or ferrimagnetic. Temperature: plot $1/\chi$ against $T$ above the critical point; it is a straight line through the origin (para), cutting the axis at $+T_c$ (ferro) or extrapolating to $-\theta$ (antiferro).

WHERE MARKS GET LOST

Diamagnetism is present in **every** material; it is just usually swamped. A substance is called diamagnetic only when it has no permanent moments.

## Dia- & paramagnetism

IN ONE LINE Diamagnetism: a field speeds up or slows down each electron orbit by the Larmor frequency $eB/2m$; the extra current opposes $B$. Paramagnetism: permanent moments turn towards $B$ against thermal disorder; the Boltzmann average gives the Langevin function and the Curie law $\chi = C/T$.

### Diamagnetism

When the flux through an orbit changes, an induced current opposes the change (Lenz's law). Switch on a field and every electron orbit is gently sped up or slowed down, in whichever sense makes a moment *against* the field. The magnetic field of that induced current is opposite to the applied field: a small, negative $\chi$. It exists in all materials, because every atom has orbiting electrons.

**The Larmor picture.** The orbit as a whole precesses about $\mathbf{B}$ at the Larmor frequency $\omega_L = eB/2m$. Each electron then carries an extra current $-Ze\,\omega_L/2\pi$, and its loop area is set by the mean square distance from the field axis, $\langle \rho^2 \rangle = \frac{2}{3}\langle r^2 \rangle$ for a spherical atom. That is where the $\langle r^2 \rangle$ and the 6 in the result come from:

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$$\chi_{\text{dia}} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2 \rangle$$

WHERE

| | |
|---|---|
| $\chi_{\text{dia}}$ | diamagnetic susceptibility |
| $\mu_0$ | permeability of free space |
| $N$ | atoms per unit volume |
| $Z$ | electrons per atom |
| $e,\ m$ | electron charge and mass |
| $\langle r^2 \rangle$ | mean square radius of the electron orbits |

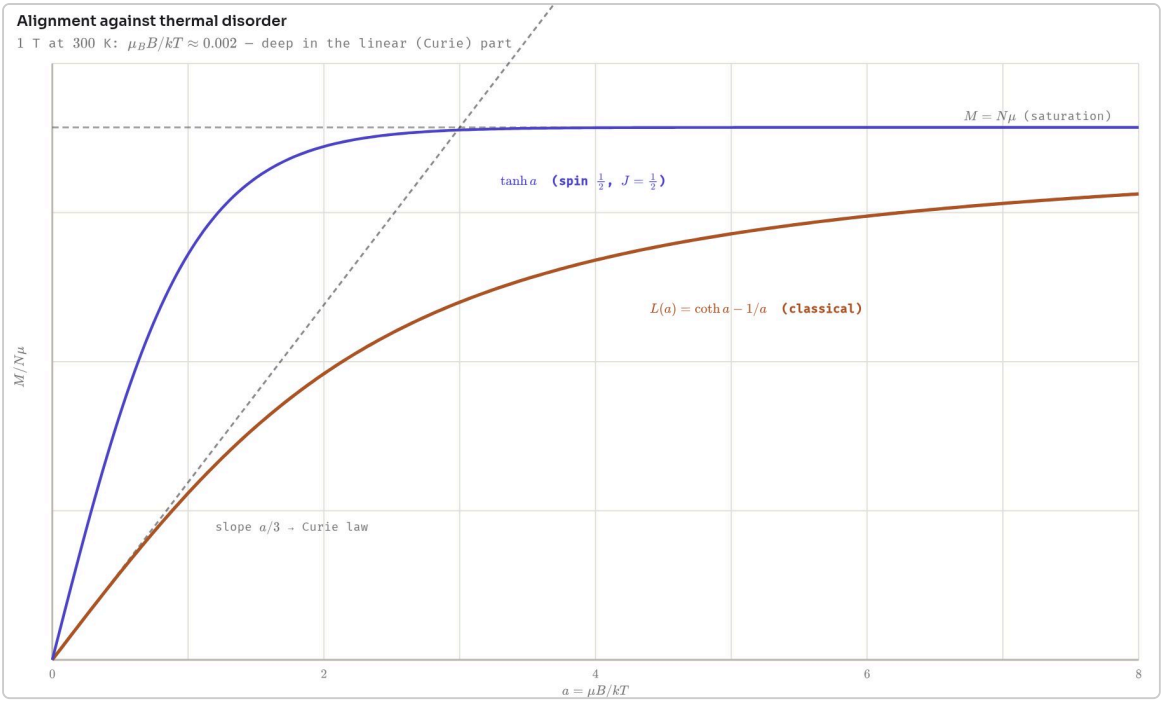
Negative, independent of temperature, and larger for big atoms with many electrons (it grows with $Z\langle r^2\rangle$). Typical size $-10^{-5}$ to $-10^{-6}$.

**Quantum theory.** With a field the Hamiltonian gains a term $\dfrac{e^2B^2}{8m}(x^2+y^2)$ (plus one in $L_z$). This energy is second order in $B$; equating it to $-\tfrac{1}{2}\chi B^2/\mu_0$ gives $\chi_{\text{dia}} = -\dfrac{\mu_0 N Z e^2}{6m}\langle r^2\rangle$, identical to the classical result, with $\langle r^2\rangle$ now taken from the wavefunction. Agreement is good for light elements and rare gases (He about $-1.9 \times 10^{-6}$).

### Paramagnetism

Now give every atom a permanent moment $\mu$. Without a field they point at random and cancel. A field tries to line them up (energy $-\mu B\cos\theta$), heat tries to scramble them ($kT$). The outcome of that contest is the Boltzmann average of $\cos\theta$. Paramagnetism is seen in:

- metals (conduction electrons);
- atoms and molecules with an odd number of electrons (free Na atoms, NO gas);
- a few with an even number ($O_2$);
- atoms and ions with a partly filled inner shell: rare earths, actinides, transition-metal ions such as $Mn^{2+}$.




The Langevin function $L(a) = \coth a - 1/a$ (classical moments, any direction allowed) and $\tanh a$ (spin $\frac{1}{2}$, only two directions). Both start as straight lines and level off at full alignment. Ordinary fields at room temperature give $a \approx 0.002$, far down on the straight start: that straight start is the Curie law.

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Curie law $\chi = C/T$, $C = \mu_0 N\mu^2/3k$, for $\mu B \ll kT$. Saturation $M_s = N\mu$ when $\mu B \gg kT$ (very low temperature, very strong field).

### The quantum theory

Quantum mechanics allows only $2J + 1$ orientations of the moment, with energies $m_J g \mu_B B$, $m_J = -J, \ldots, J$. Averaging over those few levels instead of over all angles replaces $L(a)$ by the **Brillouin function** $B_J(x)$, which tends to $L(a)$ as $J \to \infty$ and to $\tanh x$ for $J = \frac{1}{2}$. In small fields the result is again the Curie law, with the classical $\mu^2$ replaced:

$p_{\text{eff}} = g\sqrt{J(J+1)}$ is the **effective Bohr magneton number**, the number the experiments measure. $g$ is the Landé factor, $g = 1 + \dfrac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$.

**Hund's rules** (ground state of an incomplete shell): (i) spins add to the largest $S$ allowed by Pauli; (ii) then the largest $L$; (iii) $J = |L - S|$ if the shell is less than half full, $J = L + S$ if more than half full; half full gives $L = 0$, $J = S$.

**Quenching.** In iron-group salts the crystal field of the neighbouring ions locks the orbits in place, so $L$ contributes almost nothing: use $J = S$, $g = 2$. Rare-earth 4f shells lie deep inside the ion, are shielded, and keep the full $J$. (Problems on $Cr^{2+}$, $Cu^{2+}$ and $Ni^{2+}$ all use quenching.)

WHERE MARKS GET LOST

The quantum result uses $\mu^2 = p_{\text{eff}}^2 \mu_B^2 = g^2 J(J+1)\mu_B^2$, not $g^2 J^2 \mu_B^2$. The difference is the "$+1$".

## Ferromagnetism

IN ONE LINE Weiss: an internal field proportional to $M$ lines the moments up. It explains spontaneous magnetisation below $T_c$ and the Curie–Weiss law above it. The field is the quantum exchange interaction. A lump of iron can be unmagnetised because it is split into domains; moving and turning them gives the hysteresis loop.

### Ferromagnetism

Permanent moments, as in paramagnets, but neighbours line up **parallel even without a field**: **spontaneous magnetisation**, below the **Curie temperature** $T_c$. Above $T_c$ thermal agitation wins and the material is paramagnetic. Large, positive, field-dependent $\chi$; magnetisation against field traces a hysteresis curve. Examples: Fe ($T_c = 1043$ K), Ni (631 K), Co (1388 K), Gd (292 K), Dy; alloys and oxides such as MnBi, MnAs, $CrO_2$.

Two puzzles need explaining. Why do the moments line up at all, when a paramagnet needs an enormous field to do the same? And if iron is spontaneously magnetised, why is an ordinary iron nail not a magnet? Weiss answered both with two hypotheses.

### Weiss theory

Two hypotheses: (i) the specimen is made of small spontaneously magnetised **domains**, whose moments add as vectors; (ii) inside a domain an **exchange (molecular) field** $B_E = \lambda M$ lines the moments up, $\lambda$ the Weiss constant (independent of $T$). For iron $B_E \approx 1000$ T.

**Above $T_c$:** treat the material as a paramagnet that feels $B + \lambda M$ instead of $B$. Solving for $M$ gives the Curie–Weiss law:

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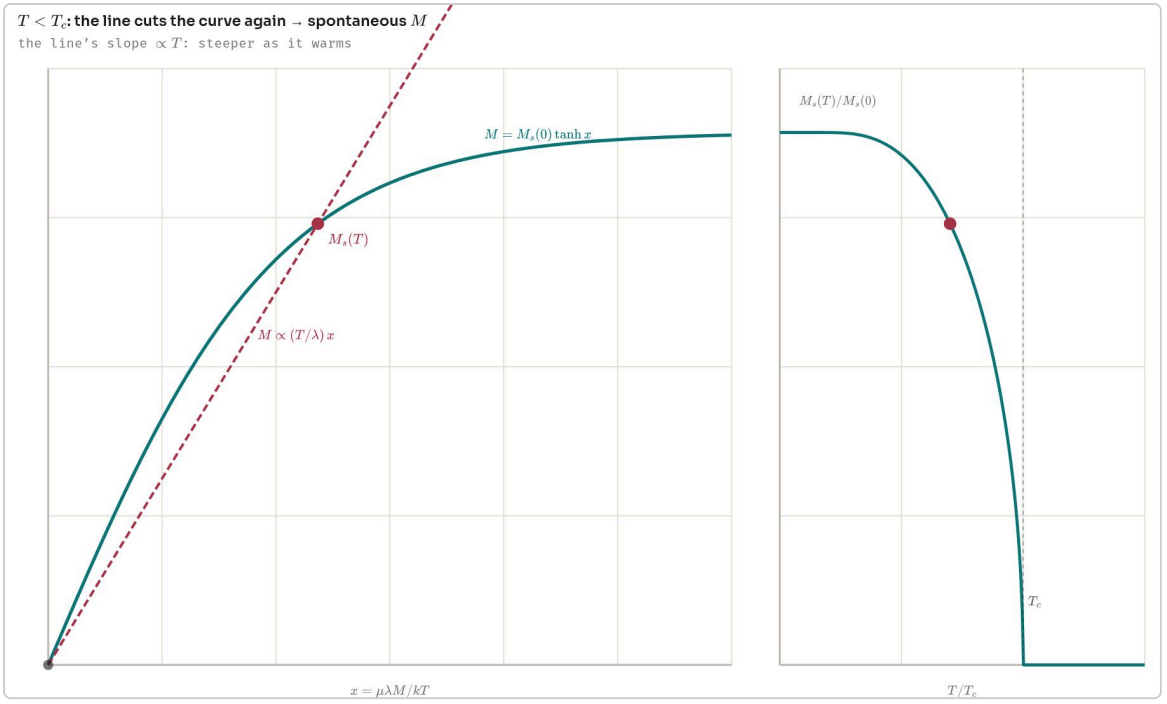
$$\chi = \frac{C}{T - T_c}, \qquad T_c = \frac{\lambda C}{\mu_0}, \qquad C = \frac{\mu_0 N g^2 J(J+1)\mu_B^2}{3k}$$

WHERE

| | |
|---|---|
| $\chi$ | susceptibility above the Curie point |
| $C$ | Curie constant |
| $T$ | temperature |
| $T_c$ | Curie temperature |
| $\lambda$ | molecular-field (Weiss) constant |
| $N$ | atoms per unit volume |
| $g$ | Landé factor |
| $J$ | total angular momentum quantum number |
| $\mu_B$ | Bohr magneton |
| $k$ | Boltzmann's constant |
| $\mu_0$ | permeability of free space |

A strong molecular field (large $\lambda$) means a high Curie temperature. Internal field from measured numbers: $B_E = \lambda M_s = \dfrac{3kT_c}{g(J+1)\mu_B}$ (quantum) or $\dfrac{3kT_c}{\mu}$ (classical).

**Below $T_c$: spontaneous magnetisation.** With $B = 0$, the magnetisation must satisfy two equations at once. The paramagnet formula says $M = M_s(0) B_J(x)$ with $x = gJ\mu_B \lambda M / kT$; turned round, the definition of $x$ says $M$ is proportional to $x$, with a slope proportional to $T$. Plot both and look for crossings:




The graphical solution (standard construction) for spin $\frac{1}{2}$. Cold: the straight line is shallow and cuts the curve again away from the origin: a non-zero magnetisation exists with no field. Warming steepens the line; at $T_c$ it is tangent at the origin; above it, only $M = 0$ is left. Right: the crossing height against $T$, the spontaneous magnetisation curve, $M_s(0)$ at 0 K and zero at $T_c$.

The fall to zero at $T_c$ is continuous, with no latent heat: a **second-order phase transition**.

### Origin of the Weiss field: exchange

A dipole–dipole field is about $10^3$ G (0.1 T), far too weak: iron needs about $10^7$ G. Heisenberg (1928): the field comes from the quantum **exchange interaction**, a consequence of the Pauli principle. Turning one spin changes the allowed spatial wavefunction and so the charge distribution: the energy is electrostatic, not magnetic. The energy of two atoms contains

$$U = -2J_e\, \mathbf{S}_i \cdot \mathbf{S}_j$$

WHERE

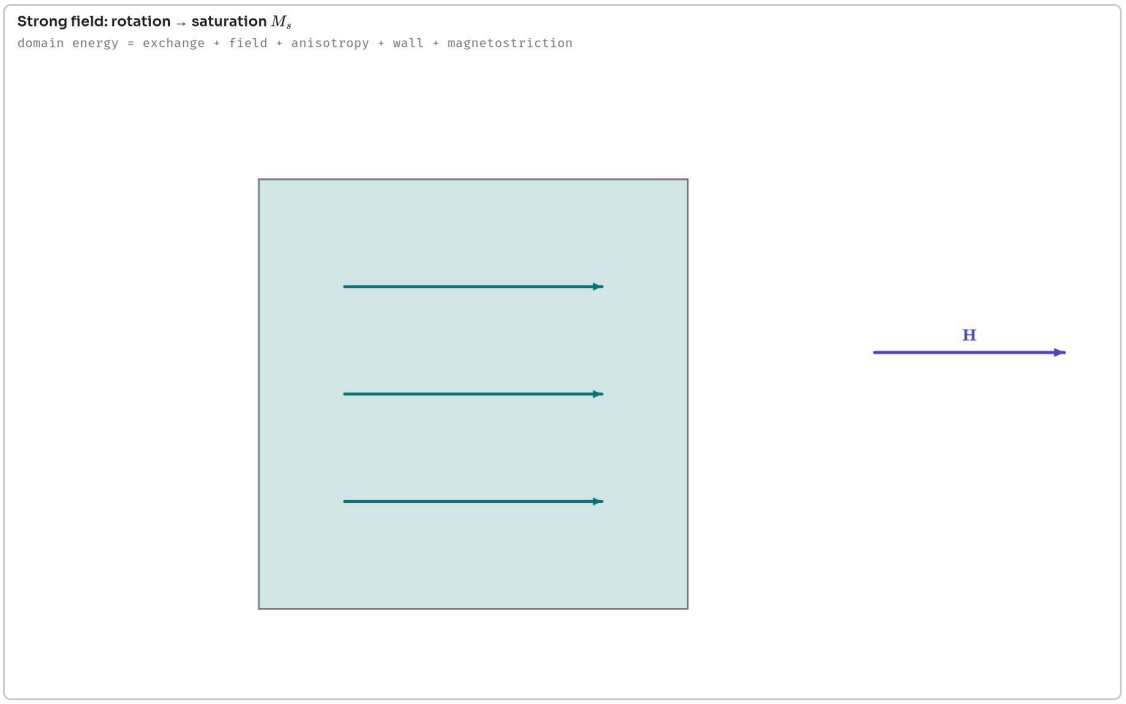
| | |
|---|---|
| $U$ | exchange energy of a pair of neighbouring atoms |
| $J_e$ | exchange integral ($J_e > 0$ favours parallel spins) |
| $\mathbf{S}_i,\ \mathbf{S}_j$ | spin vectors of the two atoms |

$J_e$, the exchange integral, depends on the overlap of the two atoms' charge clouds. $J_e > 0$: parallel spins have lower energy (ferromagnetism). $J_e < 0$: antiparallel (antiferromagnetism). Bethe–Slater: $J_e > 0$ when $r_{ab}/r_{3d}$ (atomic separation ÷ radius of the 3d shell) is a little above 3: Fe, Co, Ni, Gd are ferromagnetic; Mn and Cr are not. Summing over $z$ nearest neighbours (Stoner) gives $B_E = \lambda M$ with

$$\lambda = \frac{2zJ_e}{Ng^2\mu_B^2}.$$

### Domains and hysteresis

Why iron is usually unmagnetised: each domain is fully magnetised but the domains point in different directions. Néel: the domain pattern minimises the total energy: exchange, magnetic field energy, anisotropy energy, wall energy and magnetostrictive energy. One big domain has a large external field; splitting lowers that, but every wall costs energy, so the split stops at a balance.




Domains from the energy argument, then the two ways a field magnetises them. Steps 1–3: splitting and closure domains remove the stray field. Step 4: wall movement (low fields). Step 5: rotation (high fields), up to saturation.

Two processes magnetise a specimen: (a) **growth** of favourably oriented domains (wall movement) and (b) **rotation** of domain magnetisation towards the field. They map onto the magnetisation curve:

- **OA** (small fields): reversible wall movement.
- **AB**: large, irreversible wall movements; $M$ rises steeply.
- **BC**: domains rotate from their easy directions to the field; $M$ creeps up to saturation $M_s$.
- Reduce the field: $M$ does not retrace the curve. At $H = 0$ the **remanence** $M_r$ is left. The reverse field that removes it is the **coercivity** $H_c$. The closed loop is the **hysteresis loop**; its area is the energy lost per cycle.

**Soft** materials (small $H_c$, thin loop: transformer cores, electromagnets; soft iron, Si-steel) and **hard** materials (large $H_c$ and $M_r$: permanent magnets; alnico, ferrite magnets).

## Antiferro & ferri

IN ONE LINE Negative exchange makes neighbours antiparallel. Two equal sublattices cancel (antiferromagnetism: small $\chi$ with a cusp at the Néel temperature). Two unequal ones leave a net moment (ferrimagnetism: ferrite, magnetite), which behaves outwardly like a ferromagnet.

### Antiferromagnetism

Spins of neighbouring atoms antiparallel (exchange integral negative). Think of two interpenetrating **sublattices A and B**, magnetised in opposite directions: no net magnetisation. First seen in MnO ( $T_N = 116$ K); also FeO, NiO, CoO, $\mathbf{MnF_2}$, Cr. A field gives a small magnetisation that **increases** with temperature up to the **Néel temperature** $T_N$, then falls (paramagnetic). At $T_N$, $\chi$ is not infinite but has a weak cusp.

**Néel's theory above $T_N$.** Each sublattice is a paramagnet that feels the applied field *minus* molecular fields from the other sublattice ($\beta$) and from its own ($\alpha$). The minus signs are the whole story: the neighbours fight the field instead of helping it.

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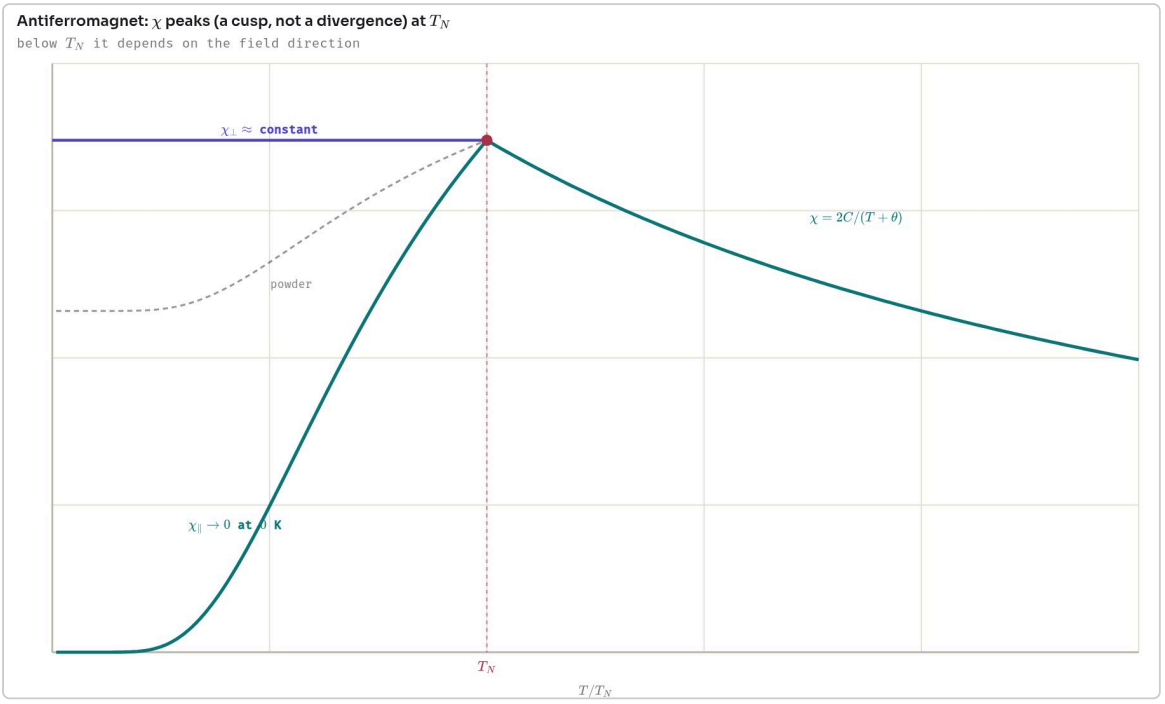
$$\chi = \frac{2C}{T + \theta} \quad (T > T_N)$$

WHERE

| | |
|---|---|
| $\chi$ | susceptibility of an antiferromagnet above $T_N$ |
| $C$ | Curie constant |
| $T$ | temperature |
| $\theta$ | constant that sets the intercept of $1/\chi$ at $-\theta$ |
| $T_N$ | Néel temperature |

Like Curie–Weiss with the sign of the intercept reversed: $1/\chi$ extrapolates to $-\theta$. In the simplest theory (only A–B interactions) $\theta = T_N$; real crystals have $\theta > T_N$ (MnO: $\theta \approx 610$ K, $T_N = 116$ K).

**Below $T_N$**: each sublattice is ordered along its preferred direction. For a field across that direction, $\chi_\perp$ stays almost constant (both sublattices simply cant a little towards the field). Along it, $\chi_\parallel$ falls to zero at 0 K: the internal field is too strong to let a perfectly ordered sublattice turn, and at 0 K both are perfectly ordered. At $T_N$, $\chi_\perp = \chi_\parallel = \chi$.




Left chip: the antiferromagnet's susceptibility from mean-field theory, normalised to its value at $T_N$. Above $T_N$, the Néel law; below, $\chi_\perp$ (indigo) and $\chi_\parallel$ (teal) split. The dashed curve is a powder, $\frac{1}{3}\chi_\parallel + \frac{2}{3}\chi_\perp$, which is what a polycrystalline sample shows: the cusp. Right chip: the ferrimagnet.

### Ferrimagnetism

The same antiparallel arrangement, but the sublattice magnetisations are **unequal**, so a net magnetisation survives. Found in **ferrites**, double oxides of the form $\mathbf{MO \cdot Fe_2O_3}$ (M = Mn, Co, Ni, Cu, Mg, Zn, Fe) with the **spinel** structure: oxygen ions in a close-packed cubic array, metal ions in the gaps between them. Two kinds of gap: **A (tetrahedral) sites** with four $\mathbf{O^{2-}}$ neighbours and **B (octahedral) sites** with six. The A–B exchange is strongly negative, so the A and B sublattices point opposite ways.

**Magnetite** $\mathbf{Fe_3O_4}$ = $\mathbf{FeO \cdot Fe_2O_3}$: one $\mathbf{Fe^{3+}}$ on an A site; one $\mathbf{Fe^{3+}}$ and the $\mathbf{Fe^{2+}}$ on B sites. The two $\mathbf{Fe^{3+}}$ moments (5 $\mu_B$ each, $3d^5$) cancel, and the net moment per formula unit is that of one $\mathbf{Fe^{2+}}$, 4 $\mu_B$ ( $3d^6$: four unpaired electrons). Experiment gives 4.1 $\mu_B$. A spinel unit cell holds 8 formula units.

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> Ferrimagnet in one line: antiparallel *and* unequal sublattices. Net moment = difference of the sublattice moments. Outwardly like a ferromagnet (spontaneous $M$, Curie point, domains, hysteresis), but with a much smaller saturation moment and a very high electrical resistivity.

Above the Curie temperature both sublattices follow Curie–Weiss-like laws and $1/\chi$ is a hyperbola rather than a straight line; below it, saturation sets in. Ferrites are ceramics with resistivities up to $10^{10}$ times that of iron, so eddy-current losses are tiny: the standard cores for high-frequency transformers, inductors, antenna rods and microwave devices.

## $P$, local field, C–M

> IN ONE LINE A field pulls the bound charges of a dielectric slightly apart: polarisation $P$. An atom feels not the average field $E$ but the local field $E + P/3\varepsilon_0$. Putting that into $P = N\alpha E_{\text{loc}}$ gives the Clausius–Mossotti relation between $\varepsilon_r$ and $\alpha$.

### Polarisation and susceptibility

Dielectrics are insulators: no free carriers, so no current flows. But their bound positive and negative charges can shift by a fraction of an atomic diameter in a field. Each atom or molecule becomes a small dipole $p$ (charge × separation), and the **polarisation** $P$ is the dipole moment per unit volume. The bound charges that appear on the surfaces make a polarisation field $E_p$ opposing the applied field:

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_p, \qquad \mathbf{P} = \varepsilon_0 \chi \mathbf{E}$$

> WHERE
>
| | |
|---|---|
| $\mathbf{E}$ | field inside the dielectric |
| $\mathbf{E_0}$ | applied field |
| $\mathbf{E}_p$ | field of the polarisation charges (it opposes $\mathbf{E_0}$) |
| $\mathbf{P}$ | polarisation: dipole moment per unit volume |
| $\varepsilon_0$ | permittivity of free space |
| $\chi$ | electric susceptibility |

$\chi$: the electric susceptibility. This is the electric twin of the magnetic story: $P$ plays the part of $M$, and $\chi$ measures how easily the material polarises.

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$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 \varepsilon_r \mathbf{E}, \qquad \varepsilon_r = 1 + \chi, \qquad P = \varepsilon_0 (\varepsilon_r - 1) E$$

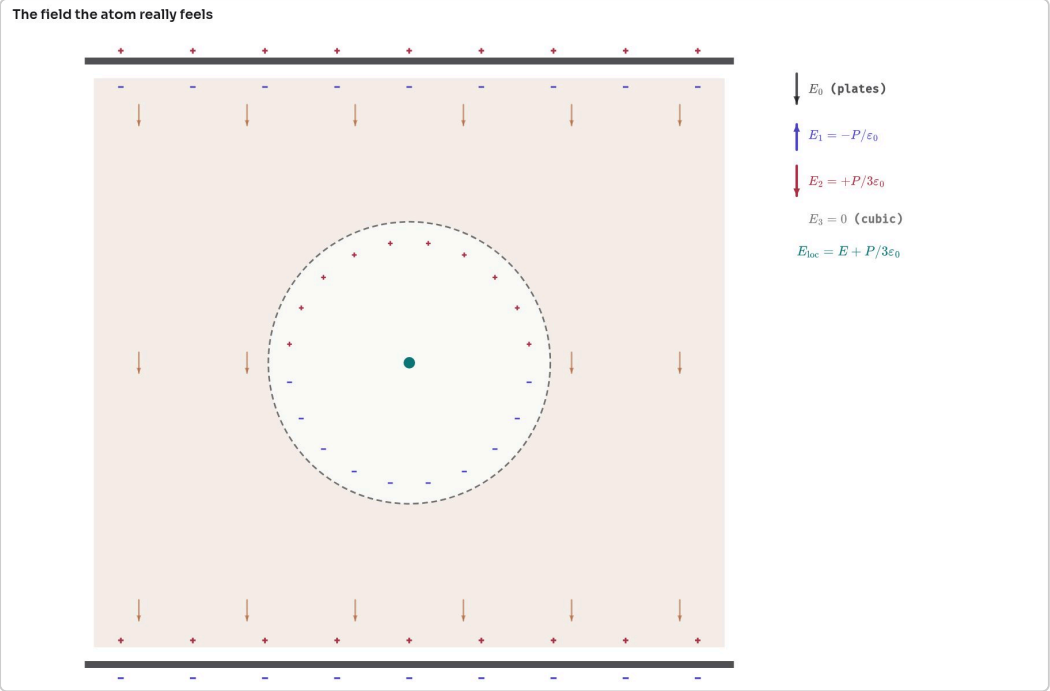
WHERE

- $\mathbf{D}$ electric displacement
- $\mathbf{E}$ electric field in the material
- $\mathbf{P}$ polarisation
- $\varepsilon_0$ permittivity of free space
- $\varepsilon_r$ relative permittivity (dielectric constant)
- $\chi$ electric susceptibility

$P$ in C/$\mathrm{m}^2$ (it equals the bound surface charge density), $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m.

### The local field

The field at the site of an atom is different from the macroscopic $E$: it includes the fields of all the other polarised atoms, and in a dense solid they are close. It is built up in four pieces, $E_{\mathrm{loc}} = E_0 + E_1 + E_2 + E_3$:




The four contributions. $E_0$ from the plates; $E_1$ from the polarisation charges on the slab's outer faces (together, the macroscopic $E$); $E_2$ from the charges on the wall of an imaginary sphere round the atom, the Lorentz cavity field; $E_3$ from the dipoles inside the sphere, zero for a cubic site.

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Lorentz relation (cubic symmetry):

$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

WHERE

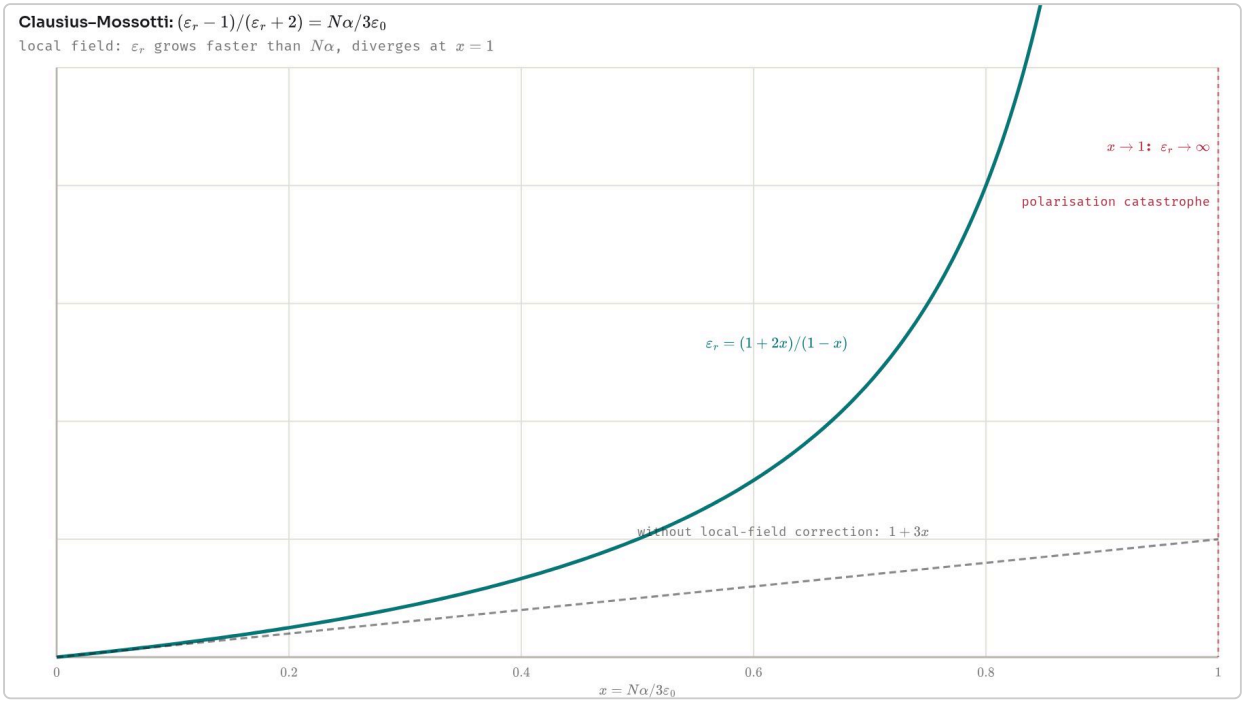
| | |
|---|---|
| $E_{\text{loc}}$ | local field acting on one atom |
| $E$ | average (macroscopic) field |
| $P$ | polarisation |
| $\varepsilon_0$ | permittivity of free space |

The atom feels *more* than the average field, because its polarised neighbours help. In a gas $P$ is tiny and $E_{\text{loc}} \approx E$.

### Dielectric constant and polarisability

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad \Longrightarrow \quad \varepsilon_r = 1 + \frac{P}{\varepsilon_0 E} = 1 + \chi$$

$\varepsilon_r$, the relative permittivity or dielectric constant, is a macroscopic property, measured with a capacitor. The **polarisability** $\alpha$, the dipole moment per unit local field ($p = \alpha E_{\text{loc}}$), is an atomic one. With $N$ atoms per unit volume, $P = N\alpha E_{\text{loc}}$ (in general $\sum_j N_j p_j$). Combining with the Lorentz field links the two:




Clausius–Mossotti as a graph. Without the local-field correction, $\varepsilon_r$ would grow in a straight line, $1 + N\alpha/\varepsilon_0$. With it, each polarised atom boosts its neighbours' field, the growth runs away, and $\varepsilon_r$ diverges as $N\alpha/3\varepsilon_0 \to 1$. That runaway is the "polarisation catastrophe" behind ferroelectricity.

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$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0} = \frac{1}{3\varepsilon_0}\sum_j N_j\alpha_j$$

WHERE

| | |
|---|---|
| $\varepsilon_r$ | relative permittivity |
| $N$ | atoms per unit volume |
| $\alpha$ | polarisability: dipole moment per unit local field |
| $N_j$, $\alpha_j$ | concentration and polarisability of species $j$ (when there are several) |
| $\varepsilon_0$ | permittivity of free space |

It connects a measured $\varepsilon_r$ to atomic polarisabilities. Exact for cubic crystals and gases; approximate for polar liquids like water.

WHERE MARKS GET LOST

Do not use $\varepsilon_r = 1 + N\alpha/\varepsilon_0$ for a solid: that ignores the local field and is only right for dilute gases (where it agrees with C–M anyway). And $\varepsilon_r$ is dimensionless, but $\alpha$ is in F $\mathrm{m}^2$ (C $\mathrm{m}^2$ $\mathrm{V}^{-1}$).

## $\alpha_e$, $\alpha_i$, $\alpha_d$

IN ONE LINE Three mechanisms add up: the electron cloud shifts ($\alpha_e$, every atom), ions shift against each other ($\alpha_i$, ionic solids), permanent dipoles turn ($\alpha_d = p^2/3kT$, polar molecules). As the frequency rises they drop out one by one: dipolar first, then ionic, leaving the electronic part at optical frequencies.

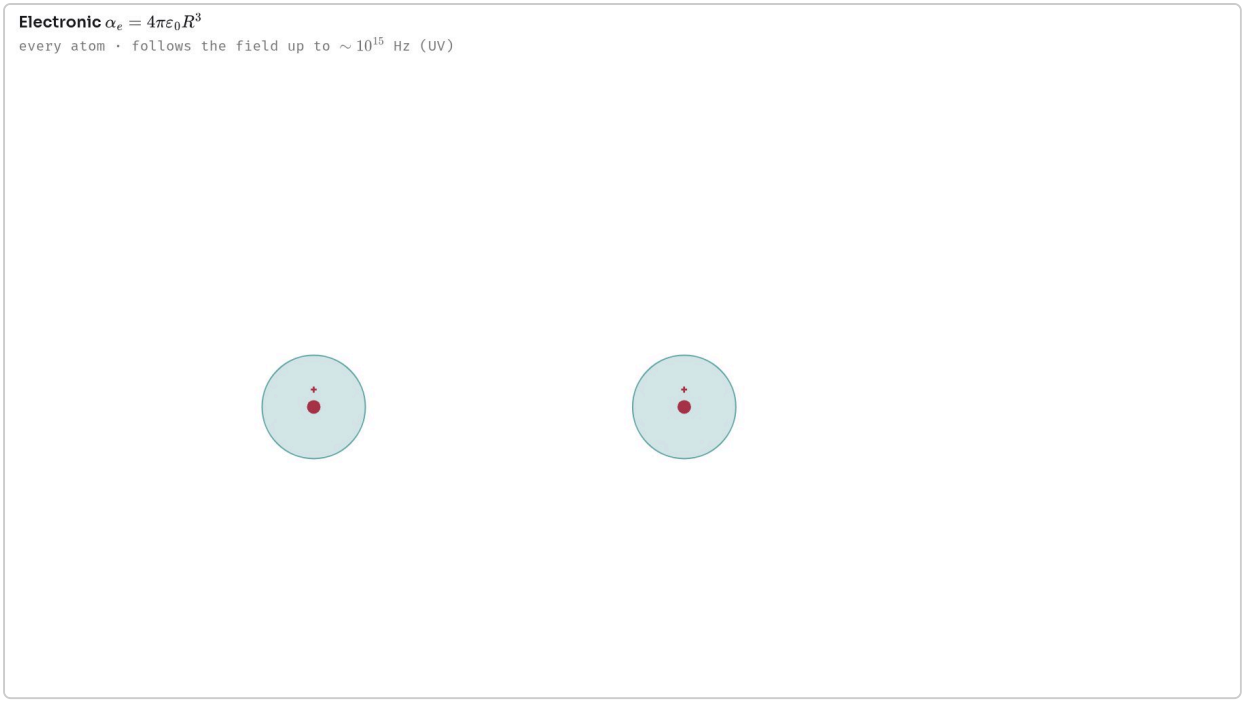
### Three ways to polarise

The total polarisability is the sum of whatever mechanisms the material allows:

$$\alpha = \alpha_e + \alpha_i + \alpha_d$$

WHERE

| | |
|---|---|
| $\alpha$ | total polarisability |
| $\alpha_e$ | electronic: electron cloud shifts against the nucleus |
| $\alpha_i$ | ionic: positive and negative ions shift |
| $\alpha_d$ | dipolar: permanent dipoles turn towards the field |




Choose a mechanism and switch the field on. Electronic: the cloud slides relative to the nucleus. Ionic: the + and − sublattices move apart. Dipolar: permanent dipoles turn partly towards the field, against thermal disorder. Every material has the first; ionic crystals add the second; polar molecules add the third.

### Electronic polarisability

The electron cloud is displaced relative to the nucleus until the restoring pull of the nucleus balances the field. Model the cloud as a uniformly charged sphere of radius $R$: the restoring force is linear in the displacement, like a spring, which gives $\alpha_e = 4\pi\varepsilon_0 R^3$, proportional to the atom's volume. The oscillator model adds the frequency:

At optical frequencies the polarisation is almost all electronic, and with $\varepsilon_r = n^2$ (Maxwell): $\dfrac{n^2-1}{n^2+2} = \dfrac{1}{3\varepsilon_0}\sum N_j \alpha_{ej}$. This is how the refractive index measures $\alpha_e$.

### Ionic polarisability

In an ionic crystal (NaCl, KCl) a field pushes the positive ions one way and the negative ions the other, stretching some bonds and compressing others. The bonds act as springs of stiffness $\beta$, so each ion pair becomes a dipole $p = e\,\Delta x$ with $\Delta x = eE/\beta$:

The ions are thousands of times heavier than electrons, so this mechanism resonates at much lower frequency (infrared, the lattice vibration frequency, $\sim 10^{13}$ Hz). It is comparable in size to $\alpha_e$: in NaCl it is about half the total.

### Dipolar polarisability

Polar molecules ($H_2O$, HCl, $NH_3$) have a permanent dipole moment $p$. Without a field they point at random; a field turns them, thermal agitation resists, and the balance gives exactly the Langevin average of paramagnetism, with $pE$ in place of $\mu B$:

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$$\alpha_e = 4\pi\varepsilon_0 R^3 \ \text{(static)}, \qquad \alpha_i = \frac{e^2}{\beta}, \qquad \alpha_d = \frac{p^2}{3kT}$$

WHERE

| | |
|---|---|
| $\alpha_e$ | electronic polarisability |
| $\alpha_i$ | ionic polarisability |
| $\alpha_d$ | dipolar polarisability |
| $\varepsilon_0$ | permittivity of free space |
| $R$ | atomic radius |
| $e$ | electron charge |
| $\beta$ | restoring-force constant of the ionic bond |
| $p$ | permanent dipole moment |
| $k$ | Boltzmann's constant |
| $T$ | temperature |

Only $\alpha_d$ depends on temperature. Plot $\alpha$ against $1/T$: slope $p^2/3k$ gives the dipole moment, intercept gives $\alpha_e + \alpha_i$ (the Debye method).

### Frequency dependence

Each process has a relaxation time. When the field changes much faster than that, the process cannot follow and stops contributing. Dipolar orientation is slowest (whole molecules turning, drops out by microwave frequencies, $\sim 10^{10}$–$10^{11}$ Hz), ionic next (infrared, $\sim 10^{13}$ Hz), electronic last (ultraviolet, $\sim 10^{15}$ Hz). So $\alpha$ and $\varepsilon_r$ fall in steps as the frequency rises. (A fourth, space-charge polarisation at interfaces, matters only at very low frequency.)

Near each step the material absorbs energy strongly (the dielectric loss peaks): this is why water absorbs microwaves and ionic crystals absorb in the infrared. Microwave ovens work at 2.45 GHz, where water's dipoles lag the field.

WHERE MARKS GET LOST

Water's static $\varepsilon_r \approx 80$ but its refractive index is 1.33 ($n^2 = 1.77$): the huge difference is the dipolar part, which is gone at optical frequencies. Never use $\varepsilon_r = n^2$ unless both are at the same frequency.

## Ferro- & piezoelectricity

IN ONE LINE A ferroelectric has spontaneous polarisation (the electric twin of ferromagnetism): a $P$–$E$ hysteresis loop, a Curie point, and $\varepsilon_r \propto 1/(T - T_c)$ above it. A piezoelectric crystal (no centre of symmetry) polarises when squeezed, and strains in a field.

### Ferroelectricity

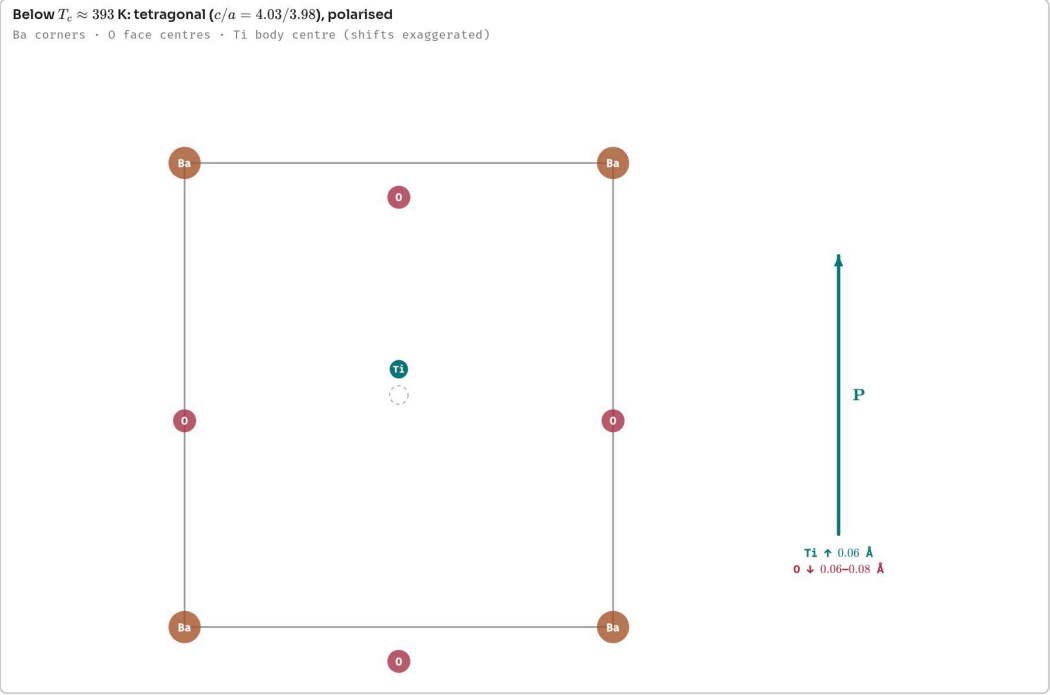
**Spontaneous polarisation**: the centres of positive and negative charge do not coincide even with no field. Everything you learnt about ferromagnets carries over with $P$ for $M$ and $E$ for $H$: domains, a **$P$–$E$ hysteresis loop** (remanent polarisation, coercive field), and a **transition temperature (Curie point)**

$T_c$ above which the crystal is paraelectric. Examples: barium titanate $\mathrm{BaTiO_3}$, Rochelle salt ( $\mathrm{NaKC_4H_4O_6 \cdot 4H_2O}$), KDP.

Two groups:

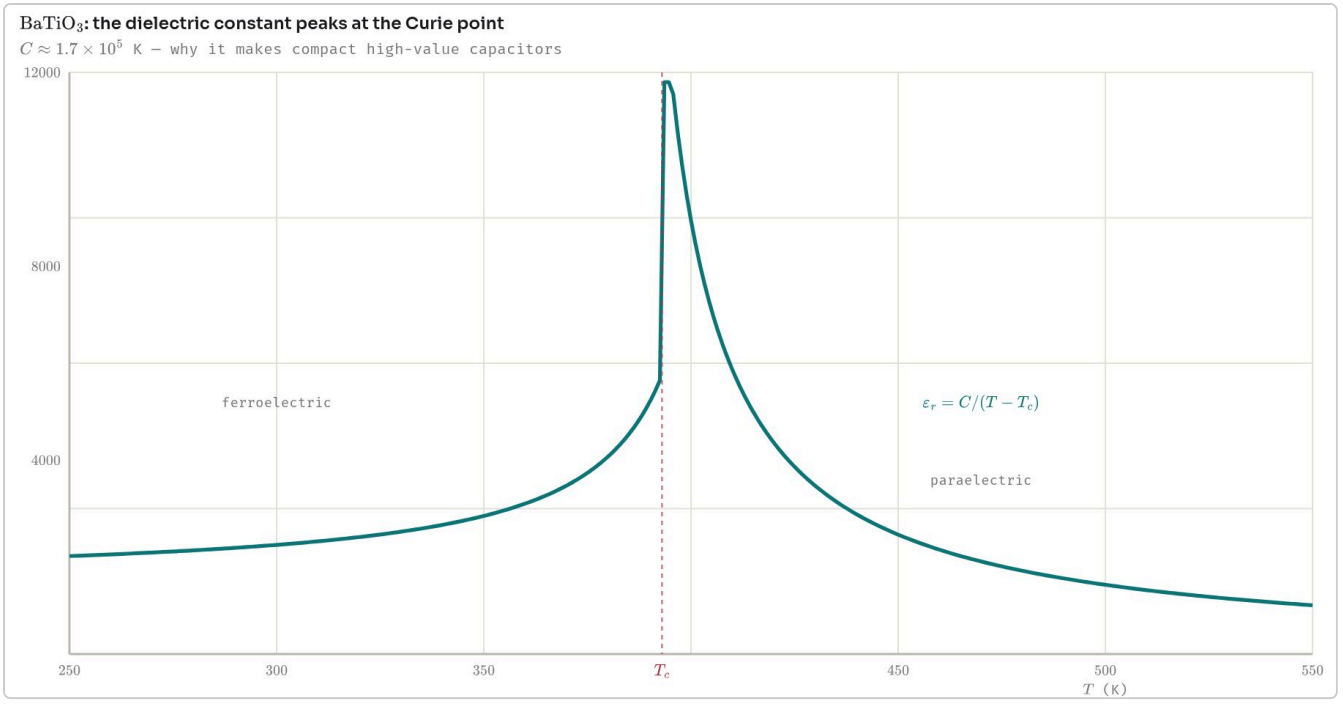
- **Order–disorder**: the transition is the ordering of individual ions, linked to protons in hydrogen bonds: $\mathrm{KH_2PO_4}$ (KDP), $\mathrm{RbH_2PO_4}$.
- **Displacive**: a whole sublattice of one kind of ion shifts relative to another; perovskite and ilmenite structures: $\mathrm{BaTiO_3}$, $\mathrm{KNbO_3}$, $\mathrm{LiTaO_3}$.

$\mathrm{BaTiO_3}$ (perovskite): cubic cell above $T_c$ (about 393 K) with $\mathrm{Ba^{2+}}$ at the corners, $\mathrm{O^{2-}}$ at the face centres and $\mathrm{Ti^{4+}}$ at the body centre (octahedrally surrounded by six $\mathrm{O^{2-}}$); the charge centres coincide. Below $T_c$, $\mathrm{Ti^{4+}}$ (and $\mathrm{Ba^{2+}}$) shift slightly up and $\mathrm{O^{2-}}$ down: the cell becomes tetragonal and polarised. The polarisation comes from induced electronic and ionic dipoles.




A face of the $\mathrm{BaTiO_3}$ cell, with the shifts exaggerated so they show. Above $T_c$ the $\mathrm{Ti^{4+}}$ sits exactly at the centre of its oxygen cage. Below $T_c$ it moves 0.06 Å up while the oxygens move 0.06–0.08 Å down, and the cell stretches along $c$: a built-in dipole in every cell.

**Why it happens: the polarisation catastrophe.** From Clausius–Mossotti, $\varepsilon_r$ diverges when $N\alpha/3\varepsilon_0$ reaches 1: the local field from the polarised neighbours alone is then enough to keep each ion polarised, so the polarisation sustains itself with no applied field.




$\varepsilon_r$ of $\mathbf{BaTiO_3}$ against temperature: a sharp peak of about $\mathbf{10^4}$ at the Curie point and the Curie–Weiss fall above it, the electric twin of $\chi = C/(T - T_c)$.

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$$\varepsilon_r = \frac{C}{T - T_c} \quad (T > T_c)$$

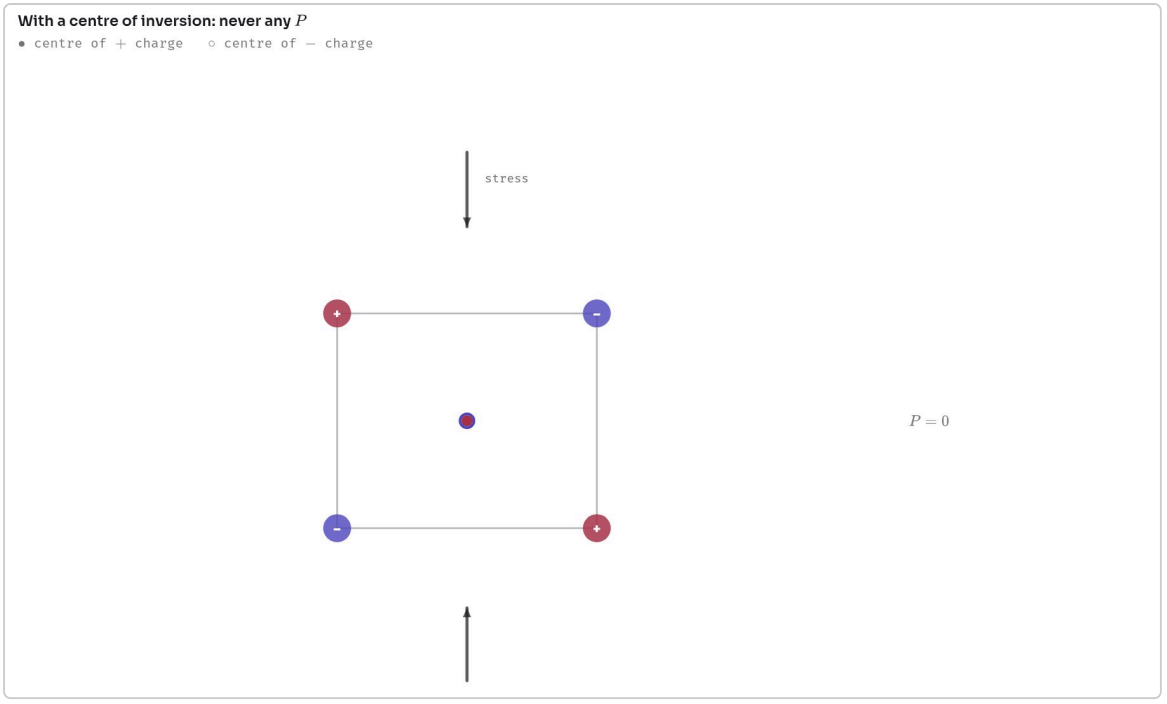
WHERE

- $\varepsilon_r$ relative permittivity
- $C$ Curie constant (of the electric case)
- $T$ temperature
- $T_c$ ferroelectric Curie temperature

Ferroelectric ↔ ferromagnetic: spontaneous $P \leftrightarrow M$, $P$–$E$ loop ↔ $M$–$H$ loop, Curie point and Curie–Weiss law in both. Uses: capacitors (very high $\varepsilon_r$ near $T_c$), non-volatile memory (FeRAM), infrared detectors.

### Piezoelectricity

In some crystals a **stress** produces a dipole moment and so surface charges (**direct effect**); a **field** produces a strain (**converse effect**). Examples: quartz, Rochelle salt, tourmaline, $\mathbf{BaTiO_3}$ and PZT ceramics.




Why the crystal must lack a centre of symmetry. In the hexagonal ring the + and − charge centres coincide until a squeeze moves the ions on the line of force unequally; then they separate and a dipole appears. In a centrosymmetric arrangement every ion has a twin opposite it, and the squeeze moves the twins symmetrically: the centres never separate.

In one-dimensional notation the two effects read:

$$P = Zd + E\varepsilon_0\chi, \qquad e = Zs + Ed$$

WHERE

- $P$ polarisation
- $Z$ stress
- $d$ piezoelectric strain constant (same in both equations)
- $E$ electric field
- $\varepsilon_0$ permittivity of free space
- $\chi$ dielectric susceptibility
- $e$ strain
- $s$ elastic compliance

$P$ polarisation, $Z$ stress, $d$ piezoelectric strain constant, $E$ field, $\chi$ dielectric susceptibility, $e$ strain, $s$ elastic compliance. The same constant $d$ appears in both equations: the direct and converse effects are two faces of one coupling. Large fields give small strains: in quartz $10^4$ V/m gives a strain of about $10^{-8}$.

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The ions must be displaced so that the charge distribution loses its symmetry: piezoelectricity needs a crystal with **no centre of inversion** (20 of the 32 point groups). Every ferroelectric is piezoelectric ($BaTiO_3$), but not every piezoelectric is ferroelectric (quartz).

Uses: electromechanical transducers: pick-ups, microphones, strain gauges, ultrasonic generators, gas lighters, and quartz oscillators that keep time in every watch and computer.